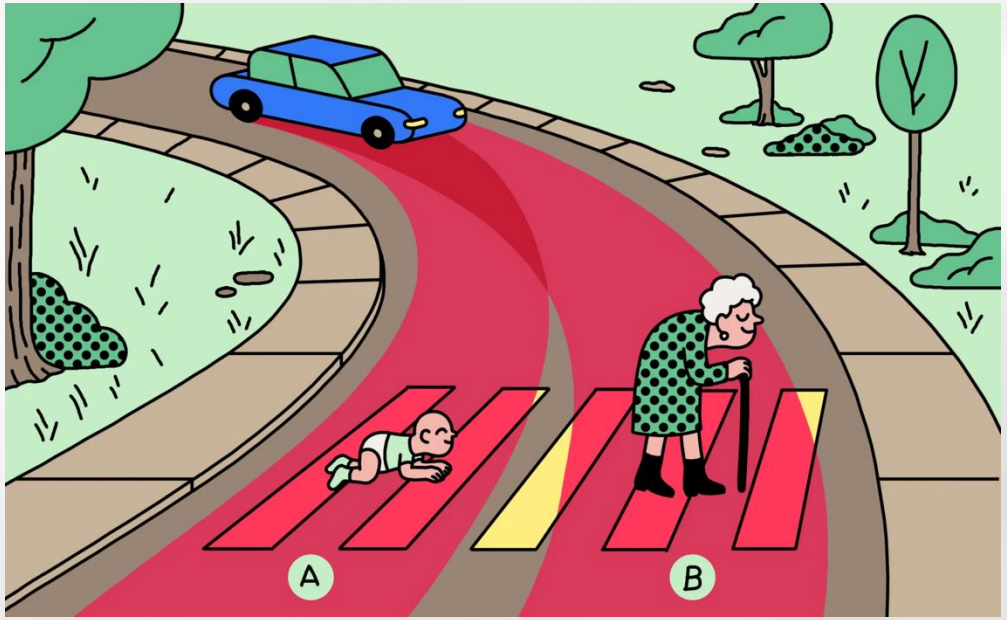
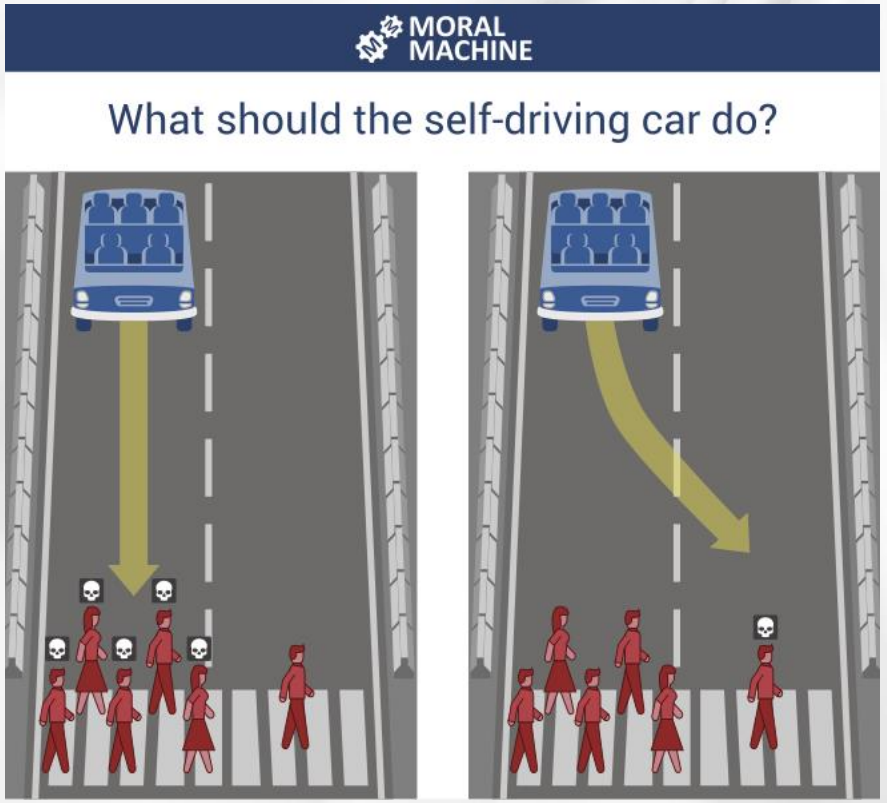
ARIN 5104:
Artificial Intelligence Ethics

Course Overview

Modern Variant of the Trolley Problem





https://www.technologyreview.com/2018/10/24/139313/a-global-ethics-study-aims-to-help-ai-solve-the-self-driving-trolley-problem/

Course Description

This course explores ethical considerations in the design, development, and deployment of artificial intelligence (AI) systems.

Students will examine the social impacts of AI and the ethical challenges that arise in areas such as privacy, bias, transparency, and accountability.

Through case studies and discussions, students will develop a framework for ethical decision-making in the development and deployment of AI systems.

Rationale

This is one of the four core courses that all students in MSc(AI) are required to take.

The rationale is multi-fold. It aims to:

- address ethical concerns of AI technologies
- prepare students for developing and deploying AI systems in the real world
- meet the growing demand in industry for AI professionals who have a strong understanding of the ethical implications of their work, and
- enhance the reputation of the MSc(AI) program by positioning itself as a leader in AI education.

Course Learning Outcomes

1. Identify ethical issues that arise in the design, development, and use of artificial intelligence systems.
2. Evaluate the fairness and bias of artificial intelligence systems and propose ways to mitigate these issues.
3. Analyze the impact of artificial intelligence on society and the workforce and propose ethical guidelines for its development and deployment.
4. Develop ethical frameworks for the governance and regulation of artificial intelligence.

Alignment with Program Learning Outcomes

1. Analyze ethical and social implications of AI technologies, and develop solutions that address issues such as bias, fairness, privacy, and accountability.
2. Demonstrate professionalism and integrity in all aspects of AI development and deployment, and adhere to ethical and legal standards.

Course Topics

Topic 1	Introduction to AI Ethics
Topic 2	Philosophical Foundations of AI Ethics
Topic 3	Bias and Fairness
Topic 4	Privacy and Data Protection
Topic 5	Transparency and Explainability
Topic 6	Accountability and Responsibility
Topic 7	Social Impact
Topic 8	Autonomous Systems
Topic 9	AI in Healthcare
Topic 10	AI Governance and Regulation
Topic 11	Future of AI Ethics

Course Assessment

Assessment	Details	Percentage
Participation (In-Class Quizzes)	Quizzes assigned via Canvas and due during class time	20%
Essay 1	Week 5; 2000 words	15%
Essay 2	Week 9; 2000 words	15%
Midterm Exam	Week 11	30%
Project	Group project with presentation	20%

Week	Date	Lecture	Assessment Assigned	Assessment Due
1	02.06	[0] Course Overview; [1] Introduction to AI Ethics		
2	02.13	[2] Philosophical Foundations of AI Ethics	Essay 1 (02.09)	
3	02.20	[3] Bias and Fairness		
4	02.27	[4] Privacy and Data Protection		
5	03.06	[5] Transparency and Explainability		Essay 1 (03.05)
6	03.13	[6] Accountability and Responsibility	Essay 2 (03.09)	
7	03.20	[7] Social Impact	Project (03.16)	
8	03.27	[8] Autonomous Systems		
9	04.03			Essay 2 (04.02)
9	04.10	[9] AI in Healthcare		
10	04.17	[10] AI Governance and Regulation		
11	04.24	Midterm Exam; [11] Future of AI Ethics	Exam (04.24)	Exam (04.24)
12	05.01			
13	05.08	Project Presentation		Project (05.07)

Main Reference Book

Bernd Carsten Stahl, Doris Schroeder, and Rowena Rodrigues (2023). Ethics of Artificial Intelligence: Case Studies and Options for Addressing Ethical Challenges. Springer.

OAPEN Open Access:
https://library.oapen.org/handle/20.500.12657/59315

Online access also available from HKUST Library website

Other References

Paula Boddington (2023). AI Ethics: A Textbook. Springer.

Franck Debos, ed. (2025). The Ethics of Artificial Intelligence. Wiley.

Online access available from HKUST Library website

Some Sources of Case Studies

Princeton Dialogues on AI and Ethics Case Studies:
https://aiethics.princeton.edu/case-studies/

Santa Clara University Case Studies (Data Science and AI):
https://www.scu.edu/ethics/focus-areas/technology-ethics/resources/embedding-ethics-into-computing-curricula-resources-and-suggestions/case-studies/

More Case Studies and Incident Registries

Accessible from AIethicist.org: https://www.aiethicist.org/ethics-cases-registries

Course Policy on Use of Generative AI Tools

While generative AI (GenAI) tools can be valuable resources for enhancing your learning experience and assisting in the search for relevant materials, all submissions for assessments must reflect your own independent work (or collaborative efforts for the group project).

Please note that plagiarism and AI checks will be conducted for all essay and project submissions to ensure academic integrity.

Failure to adhere to this policy will result in disciplinary actions.

Academic Integrity

Academic Honor Code and Academic Integrity

https://registry.hkust.edu.hk/resource-library/academic-honor-code-and-academic-integrity

Regulations for Student Academic Integrity

https://registry.hkust.edu.hk/resource-library/regulations-student-academic-integrity